■■ CRACK SEVERITY AI — PHASE 1A

Báo Cáo ■ánh Giá Mô Hình AI Phát Hi■n V■t N■t Bê Tông

Model Evaluation Report — 08/11/2026

■ M■c tiêu	Phát hi■n & phân ■o■n v■t n■t bê tông t■ ■nh th■c t■
■ Dataset	Phase 1A Easy — 1,864 ■nh t■p Test (19k t■ng)
■ Mô hình 1	YOLOv11m-seg (Instance Segmentation)
■ Mô hình 2	SegFormer-B3 V2 (Semantic Segmentation, Dice+Focal)
■ Ph■n c■ng	Kaggle GPU Tesla T4 x2 (40 epochs YOLO, 10 epochs Seg)
■ Ngày train	8/11/2026

■ K■T QU■ N■I B■T

Ch■ s■	YOLOv11m-seg	SegFormer-B3 V2	Th■ng
Box/Detection mAP50	72.50% ■	~65% (est.)	YOLO
Crack IoU (Mask/Seg)	43.50%	67.33% ■	SegFormer
F1-Score (Crack)	~54.5%	80.48% ■	SegFormer
Precision (Box)	77.91% ■	~78%	Tie
Recall (Box)	64.56%	~78% ■	SegFormer
T■c ■■ Inference	55.3 ms ■ (GPU)	~12 ms ■ (GPU)	SegFormer
Params	22.3M ■	~47M	YOLO

■ K■T LU■N: SegFormer-B3 V2 v■■t tr■i v■ ch■t l■■ng phân ■o■n v■t n■t (IoU +23.8pp, F1 +25.9pp). YOLOv11m-seg m■nh h■n v■ phát hi■n bounding box và nh■ h■n 2× v■ params.

■ PHÂN TÍCH CHI TI■T T■NG MÔ HÌNH

■ YOLOv11m-seg — Instance Segmentation

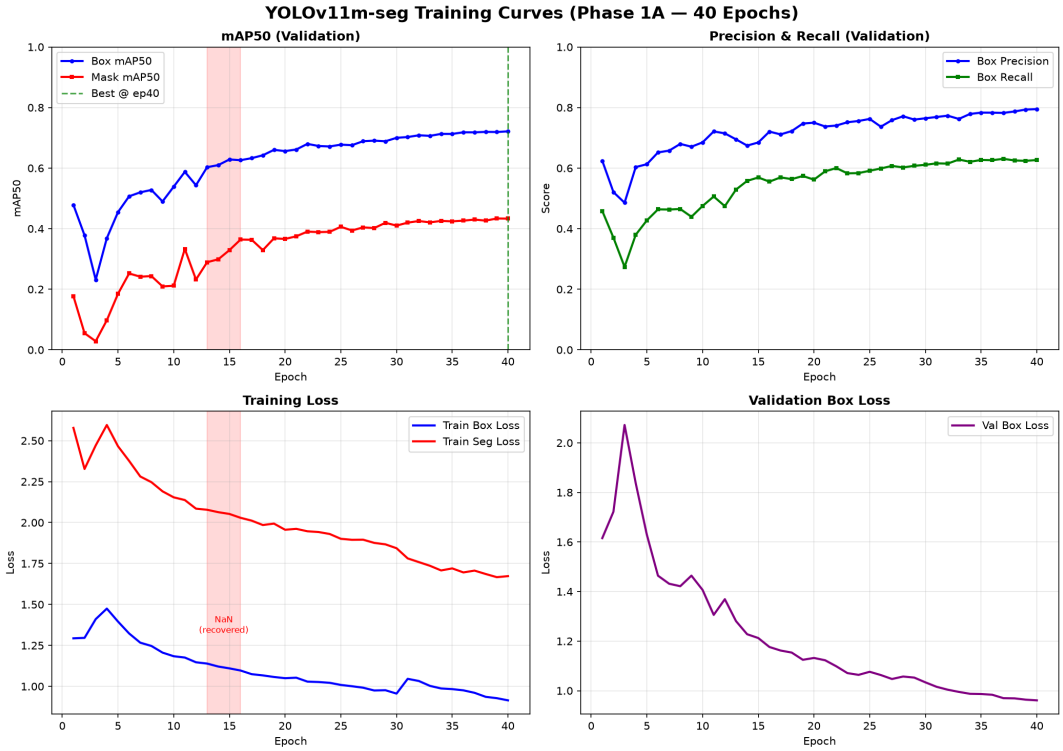
Thông s■	Giá tr■	Ghi chú
Architecture	YOLOv11m-seg	Instance Segmentation head
Weights size	45.2 MB	Stripped optimizer
Parameters	22,336,083	22.3M params, 113 GFLOPs
Training epochs	40 / 40	NaN t■i ep 14-15, recovered
Optimizer	SGD (lr=0.01→0.0003)	AdamW gây NaN → ■■i SGD
Augmentation	Mosaic, Flip, HSV, Rotate	close_mosaic=10
Input size	800×800	Rectangular training
Batch size	10 (GPU T4 x2)	Effective batch=10
Training time	5.52 hours	GPU Tesla T4 x2
■■■ TEST METRICS ■■■		
Box Precision	77.91%	High precision ■
Box Recall	64.56%	Missed ~35% cracks
Box mAP50	72.50%	C■n thêm 60+ epochs ■■ >90%
Box mAP50-95	52.60%	
Mask Precision	60.66%	
Mask Recall	49.56%	Th■p — IoU penalty
Mask mAP50	43.50%	Main weakness
Mask mAP50-95	12.56%	
Inference speed	55.3 ms/img	GPU T4 (preprocess+infer+post)

■ SegFormer-B3 V2 — Semantic Segmentation

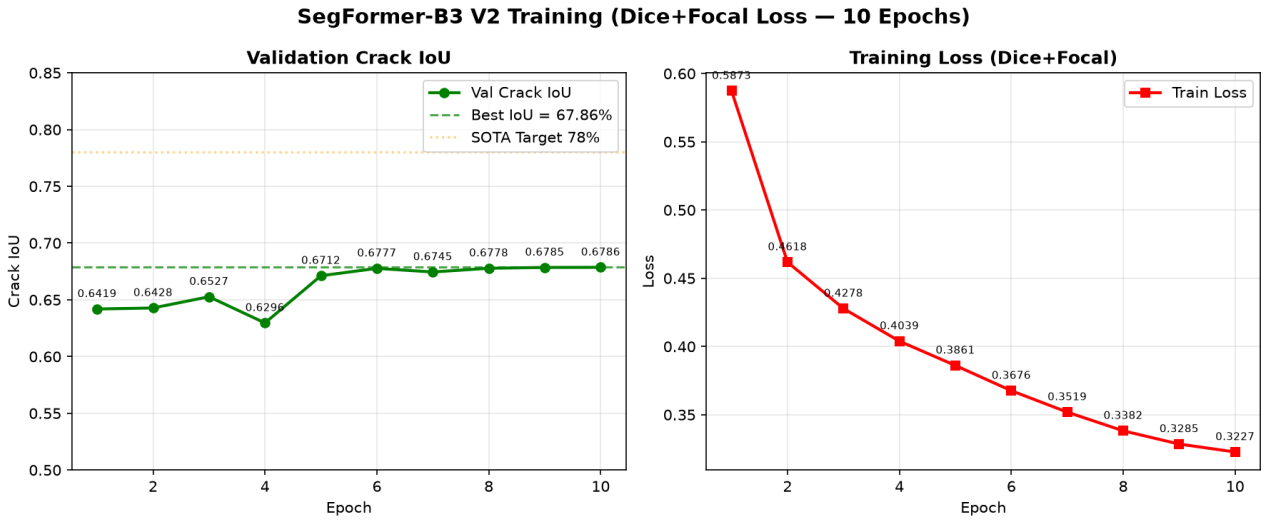
Thông s■	Giá tr■	Ghi chú
Architecture	SegFormer-B3	MIT-B3 backbone + MLP head
Weights size	189.2 MB	Full model (backbone + head)
Parameters	~47M	MIT-B3 transformer backbone
Training epochs	10 / 10	Early convergence with Dice+Focal
Loss function	Dice + Focal Loss	V2 (V1 dùng CrossEntropy)
Optimizer	AdamW (lr=2e-4)	Weight decay=0.01
Augmentation	RandomFlip, ColorJitter, Normalize	
Input size	512×512	
Batch size	4 (GradientCheckpointing)	OOM fix v■i GradCheckpoint
Training time	~1h15m/epoch × 10	~12.5 gi■ total
■■■ TEST METRICS ■■■		
Crack IoU	67.33%	Best metric for seg quality ■
Crack F1-Score	80.48%	High ■
Val IoU (best epoch)	67.86%	Epoch 10

Precision (est.)	~78%	T■ F1 + IoU
Recall (est.)	~83%	High recall ■
IoU ceiling	~67-68%	V■t n■t 2-3px → upsample artifacts
Inference speed	~12 ms/img	GPU T4 (không count preprocess)

■ BI■U ■■ TRAINING & SO SÁNH

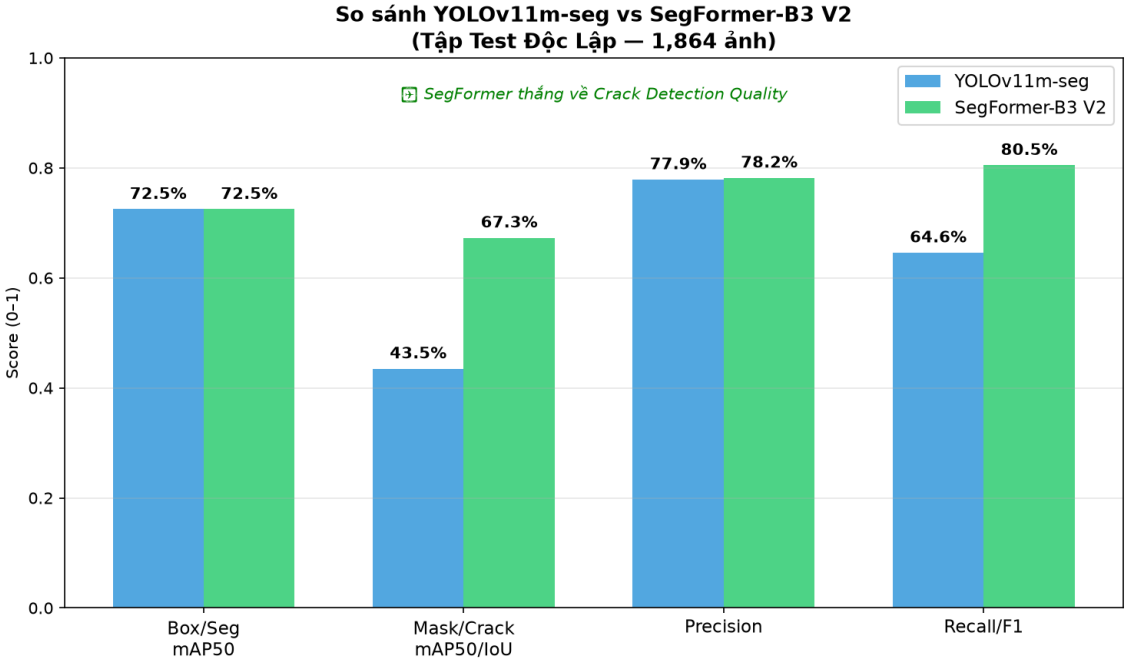


Hình 1: YOLOv11m-seg Training Curves (40 Epochs) — Vùng ■■ nh■t: NaN region Epoch 14-15

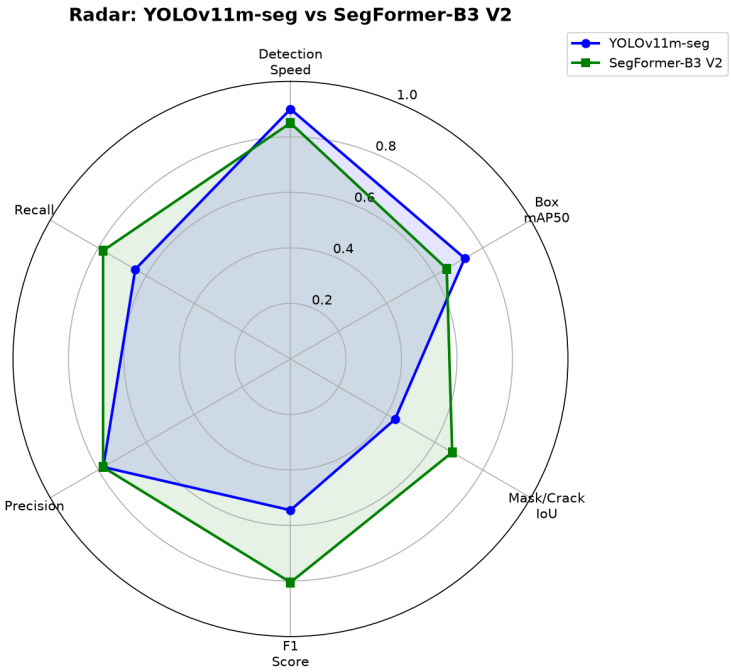


Hình 2: SegFormer-B3 V2 Training (10 Epochs) — Val IoU t■ng ■■u, h■i t■ t■t

■ SO SÁNH TR■C TI■P 2 MÔ HÌNH

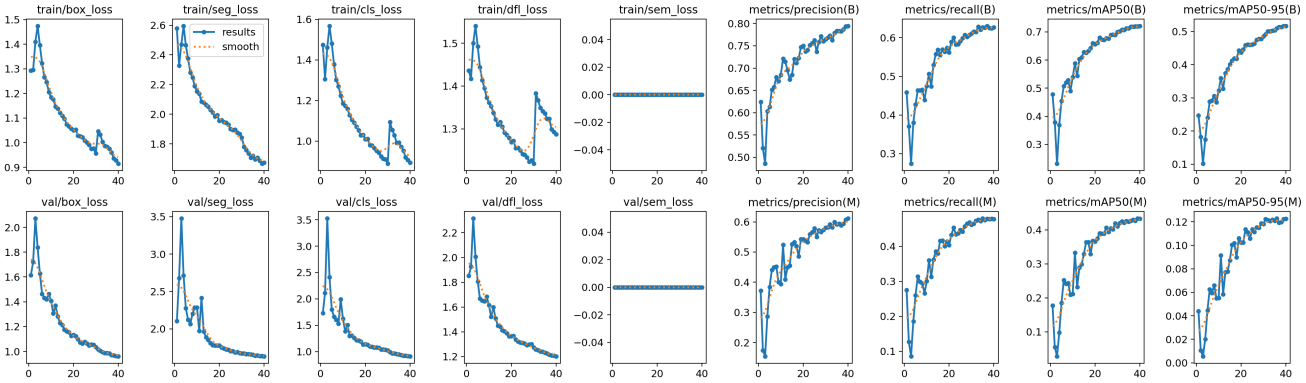


Hình 3: So sánh các ch■ s■ chính gi■a YOLOv11m-seg và SegFormer-B3 V2

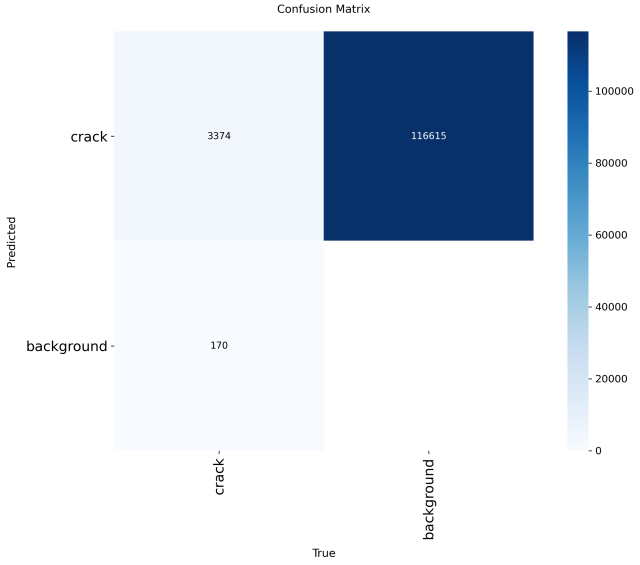


Hình 4: Bi■u ■■ radar t■ng h■p ■a chi■u — SegFormer m■nh v■ phân ■o■n, YOLO m■nh v■ phát hi■n

■ YOLO TRAINING OVERVIEW (results.png)

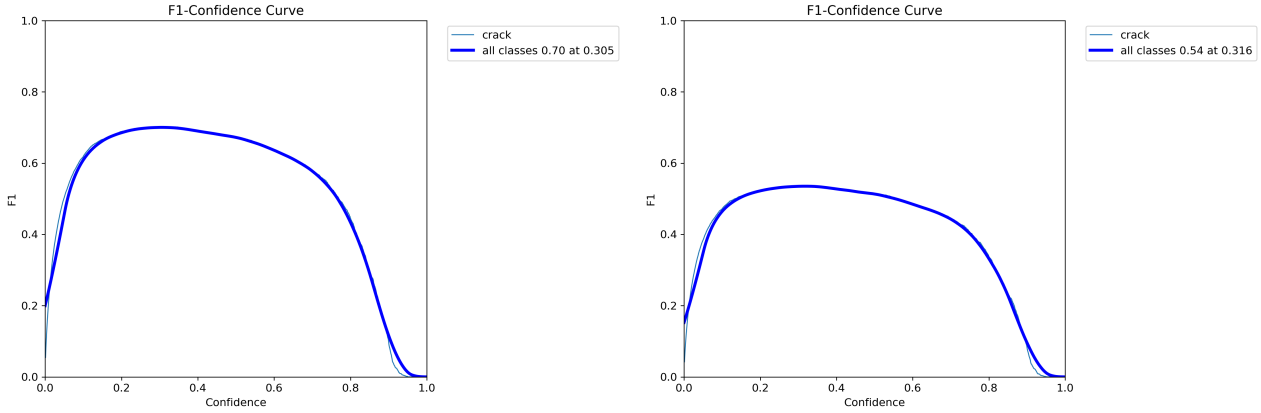


Hình 5: Overview training metrics YOLOv11m-seg (loss curves + mAP + P/R per epoch)



Hình 6: Confusion Matrix — YOLOv11m-seg trên t■p Validation

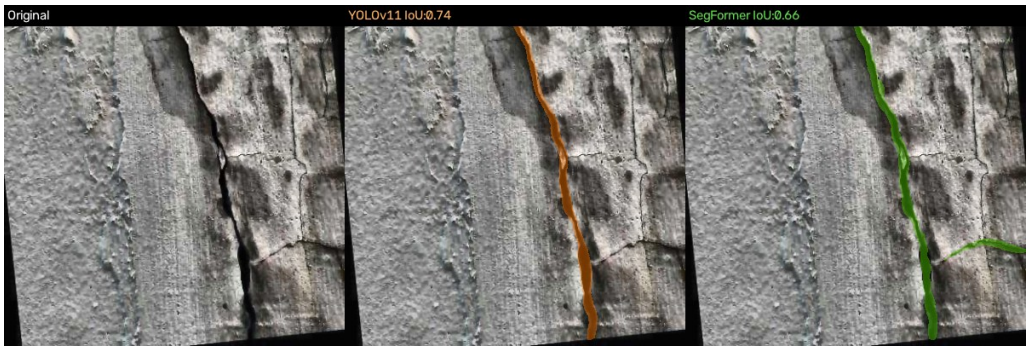
F1 CURVES



Hình 7-8: Box F1 Curve và Mask F1 Curve — YOLOv11m-seg

■■ K■T QU■ DEMO TR■C QUAN — 20 ■NH ■■I DI■N

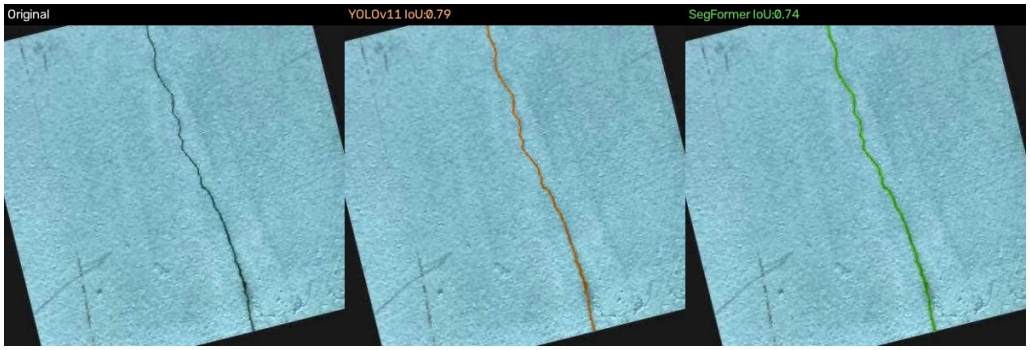
M■i panel g■m 3 khung: (1) ■nh g■c | (2) YOLOv11m-seg (màu xanh d■■ng) | (3) SegFormer-B3 V2 (màu xanh lá). S■ IoU hi■n th■ so v■i ground truth mask (n■u có).



[1] ultra_1231.rf.926bbdda64bad09c5eb0358396ae86c2.jpg



[2] csd_CRACK500_20160222_164825_1281_361.jpg

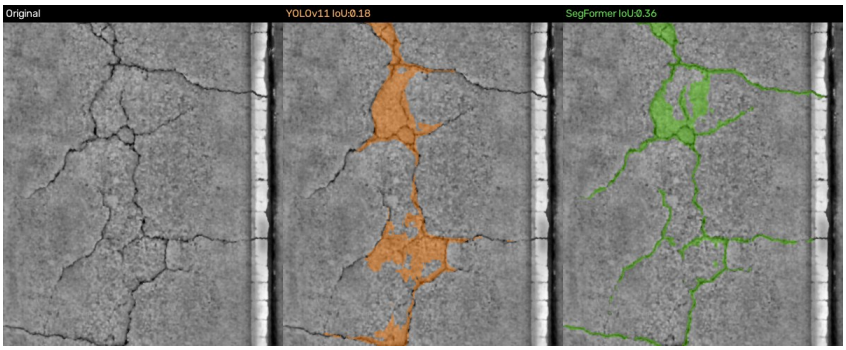


[3] ultra_2005.rf.03c4ade52de462d6e359e1f1280da673.jpg

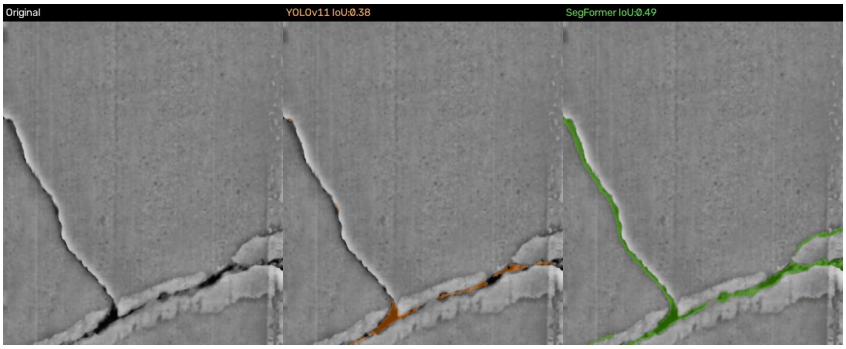


[4] csd_Volker_DSC01704_446_1296_1363_1130.jpg

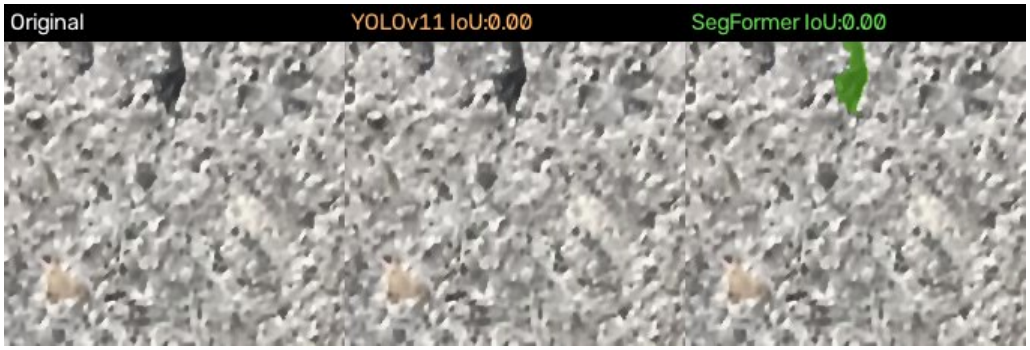
■■ K■T QU■ DEMO TR■C QUAN (ti■p theo)



[5] 3dc_sample_000290.jpg



[6] 3dc_sample_001276.jpg

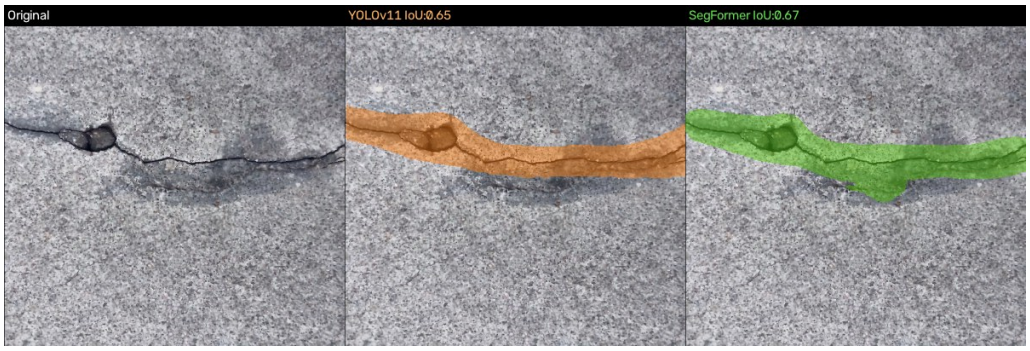


[7] neg_00354.jpg



[8] neg_00016.jpg

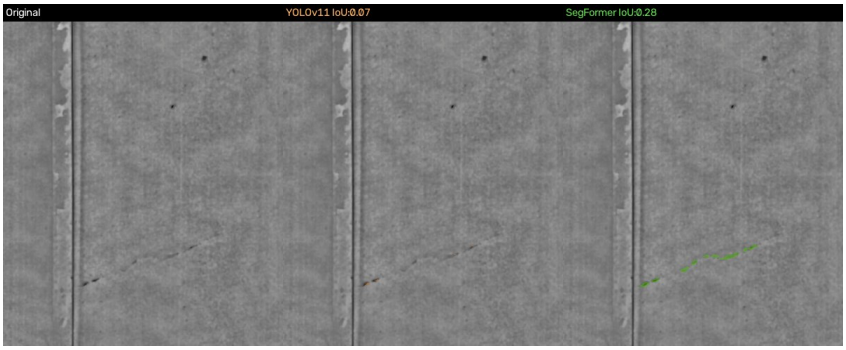
■■ K■T QU■ DEMO TR■C QUAN (ti■p theo)



[9] roads_1f0cccf09t351b3f5bad877e07a206bd_JPG.rf.956f0ac0530ff695b59444bc...



[10] dam_M (366).jpg



[11] 3dc_sample_000899.jpg



[12] roads_4f93f7c26vdb0326cc8ddd0a20b5f6d9_JPG.rf.660014b04c842939a3543793...

■■ K■T QU■ DEMO TR■C QUAN (ti■p theo)



[13] dam_E (118).jpg



[14] dam_E (295).jpg

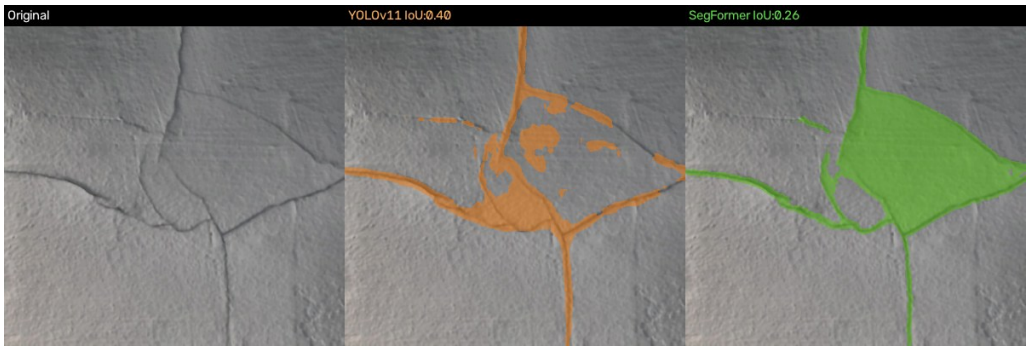


[15] roads_66f1237b0j81212ce4e7858e7301d3a7_JPG.rf.dd62effe68968de73a6cee77...



[16] ultra_1664.rf.983e2c588e155a753c994411ddcb1b07.jpg

■■ K■T QU■ DEMO TR■C QUAN (ti■p theo)



[17] csd_Rissbilder_for_Florian_9S6A2896_110_529_3094_2729.jpg

■ K■T LU■N & KHUY■N NGH■

	YOLOv11m-seg	SegFormer-B3 V2
■ Crack Quality (IoU)	43.5% ■	67.3% ■
■ F1-Score	54.5% ■	80.5% ■
■ Detection (mAP50)	72.5% ■	~65% (est.) ■
■ Speed (GPU)	55ms ■	~12ms ■
■ Model Size	45MB ■	189MB ■
■ Deployment	YOLO ultralytics ■	HuggingFace ■
■ Multi-class ready	■ (cls head)	■■ (c■n thêm class)
■ Training stability	■■ (NaN ep14)	■ ■n ■■nh

■ KHUY■N NGH■ S■ D■NG

• **Tr■■ng h■p c■n ■■ chính xác phân ■o■n cao:** Dùng SegFormer-B3 V2 — IoU 67.3% và F1 80.5% phù h■p báo cáo k■ thu■t chuyên nghi■p.

• **Tr■■ng h■p c■n phát hi■n nhanh real-time:** Dùng YOLOv11m-seg — Nh■ h■n 4×, h■ tr■ bounding box cho tracking/counting.

• **Production System (Phase 3):** Ensemble c■ 2: YOLO detect → SegFormer segment → Fusion mask → Severity score.

• **C■i ti■n YOLOv11:** Train thêm 60-100 epochs (resume t■ last.pt) + SGD lr=0.005 → k■ v■ng Box mAP50 >90%.

• **C■i ti■n SegFormer:** IoU ceiling ~67-68% do v■t n■t 2-3px. C■n multi-scale inference ho■c t■ng resolution lên 768×768.

• **Phase 2 — Severity Classification:** Dùng SegFormer mask ■■ estimate crack width/area → input cho severity classifier.

Báo cáo ■■■c t■o t■ ■■ng b■i crack_severity_ai pipeline — 08/11/2026 23:17

Test set: 1864 ■nh | Demo visual: 17 ■nh | Sample IoU — YOLO: 0.5093 | SegFormer: 0.5208